

LAJES AP, Portugal

WMO: 085090

Lat: 38.768N

Lon: 27.087W

Elev: 54

StdP: 100.68

Time Zone: -1.00 (AZR)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.9	9.9	4.0	5.1	11.4	5.0	5.4	11.8	18.2	15.0	15.5	14.4	2.7	320	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	27.2	22.3	26.2	21.8	25.7	21.5	23.1	25.9	22.6	25.2	22.1	24.6	5.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	16.9	24.5	21.8	16.6	24.2	21.1	15.9	23.7	68.5	25.9	66.5	25.3	64.6	24.6	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.9	12.0	10.4	DB	6.3	29.8	1.0	0.9	5.6	30.4	5.1	30.9	4.6	31.4	3.9	32.1
			WB	5.5	24.4	1.0	0.8	4.7	24.9	4.1	25.4	3.6	25.9	2.8	26.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	14.8	14.5	14.9	15.6	17.1	19.4	21.7	22.7	21.7	19.4	17.0	15.4
	DBStd	3.39	1.84	1.84	1.78	1.74	1.73	1.74	1.66	1.52	1.65	1.97	1.92	2.02
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	612	111	106	107	82	46	8	0	0	1	11	49	92
	CDD10.0	2878	148	127	152	169	222	281	362	395	352	292	209	168
	CDD18.3	448	0	0	1	2	8	40	104	137	102	45	8	2
	CDH23.3	1090	0	0	0	1	3	42	265	499	238	41	1	0
	CDH26.7	66	0	0	0	0	0	1	16	39	11	0	0	0
Wind	WSAvg	4.7	5.5	5.3	5.4	5.1	4.7	4.2	3.7	3.6	3.9	4.9	5.1	5.4
Precipitation	PrecAvg	1154	161	137	131	76	46	47	30	49	67	110	115	182
	PrecMax	1518	357	402	283	137	142	105	69	120	153	249	276	421
	PrecMin	669	38	10	19	8	12	18	2	5	18	32	16	16
	PrecStd	226	86	103	70	46	33	26	20	27	37	66	72	120
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.0	20.1	21.0	21.9	23.7	26.0	28.1	29.1	28.0	25.8	23.1	20.9
		MCWB	16.7	16.1	16.8	17.1	18.8	21.2	22.6	23.2	22.5	21.2	19.6	17.8
	2%	DB	19.0	19.0	19.6	20.2	22.1	24.3	26.9	27.8	26.7	24.3	21.8	20.0
		MCWB	16.5	15.8	15.8	16.5	18.0	20.2	21.9	22.6	22.1	20.6	18.7	17.5
	5%	DB	18.2	18.1	18.6	19.2	21.1	23.4	25.9	26.8	25.8	23.3	20.9	19.1
		MCWB	16.1	15.5	15.4	15.8	17.5	19.6	21.4	22.1	21.5	20.1	18.4	17.1
	10%	DB	17.8	17.2	17.8	18.6	20.1	22.4	24.9	25.9	24.8	22.6	20.0	18.5
		MCWB	15.8	15.0	15.1	15.4	16.8	18.9	20.9	21.7	20.9	19.7	17.8	16.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.7	17.3	17.6	18.3	19.7	21.9	23.5	24.2	23.5	22.2	20.4	18.7
		MCDB	18.7	18.7	19.7	20.4	22.0	24.4	26.8	27.3	25.9	24.1	22.2	19.8
	2%	WB	17.1	16.6	16.6	17.3	18.7	21.0	22.6	23.4	22.7	21.5	19.5	18.1
		MCDB	18.3	18.1	18.5	19.2	21.1	23.4	25.3	26.2	25.2	23.4	21.0	19.3
	5%	WB	16.6	16.0	16.0	16.6	18.1	20.3	22.1	22.8	22.2	20.9	18.8	17.5
		MCDB	17.9	17.5	17.8	18.5	20.4	22.7	24.6	25.5	24.6	22.7	20.2	18.8
	10%	WB	16.0	15.4	15.5	16.0	17.5	19.5	21.5	22.3	21.6	20.3	18.2	16.8
		MCDB	17.4	17.0	17.3	17.9	19.7	21.9	24.0	24.9	23.9	22.1	19.7	18.2
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.0	5.0	5.2	5.4	5.6	5.7	5.9	5.5	5.1	4.5	4.5
		MCDBR	4.8	5.6	5.8	6.2	6.5	6.6	6.9	7.2	6.8	6.0	5.4	4.9
		MCWBR	3.9	4.1	3.8	4.0	3.9	3.8	3.7	3.8	3.7	3.9	3.9	3.9
	5% WB	MCDBR	4.4	5.1	5.2	5.4	5.9	6.1	6.0	6.1	5.7	5.3	4.5	4.2
		MCWBR	3.9	4.3	3.8	3.9	3.9	4.0	3.6	3.7	3.6	3.9	3.8	3.9
Clear-Sky Solar Irradiance	taub	0.362	0.370	0.400	0.398	0.390	0.390	0.381	0.374	0.374	0.379	0.363	0.363	
	taud	2.456	2.439	2.347	2.379	2.439	2.473	2.509	2.551	2.552	2.507	2.485	2.428	
	Ebn at Noon	804	850	857	880	891	888	892	893	875	830	797	769	
	Edh at Noon	83	96	115	118	113	110	105	98	93	88	80	80	
All-Sky Solar Radiation	RadAvg	2.01	2.88	4.02	5.12	6.01	6.33	6.44	6.11	4.85	3.32	2.26	1.71	
	RadStd	0.16	0.16	0.28	0.42	0.40	0.44	0.35	0.23	0.25	0.17	0.12	0.11	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+1.02	N/A	+0.35	+0.52	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	+0.57	N/A	+0.28	+0.43	N/A	N/A	N/A	N/A

Nomenclature: See separate page